UNITED STATES INTERNATIONAL UNIVERSITY – AFRICA

School of Science & Technology · Department of Data Science and Analytics

PILOT TEST SCIENTIFIC REPORT

Predicting Maternal Health Risk Among Pregnant Women in Kenya

Using Machine Learning and Bayesian Inference on the Kenya Demographic and Health Survey 2022

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This document constitutes the pilot test report for the DSA 3900 Final Year Research Project. All models were trained on simulated data ($n = 1{,}000$) mirroring the KDHS 2022 demographic and health survey structure.

Abstract

Maternal mortality remains a critical public health challenge in Kenya, with an estimated 355 deaths per 100,000 live births—five times the Sustainable Development Goal (SDG) 3.1 target of 70 per 100,000. Despite the deployment of over 100,000 Community Health Workers (CHWs) across all 47 Kenyan counties, no data-driven, individual-level maternal health risk prediction tool currently exists to support clinical decision-making at the community level.

This pilot test report documents the end-to-end validation of a machine learning pipeline using 1,000 simulated records mirroring the statistical structure of the Kenya Demographic and Health Survey 2022 (KDHS 2022). The pipeline comprises a four-tier model architecture: Logistic Regression and Decision Tree (Tier 1 baselines); Random Forest, XGBoost, and CatBoost (Tier 2 classical ensembles); TabNet and TabTransformer (Tier 3 novel deep learning architectures); and a weighted Stacking Ensemble (Tier 4 final model). Bayesian inference via PyMC with the No-U-Turn Sampler (NUTS) augments predictions with posterior probability distributions and 95% Highest Density Intervals (HDI). SHAP values provide model-level and individual-level explainability. The final deliverable is a Streamlit web application with AI API integration, generating plain-language risk explanations and referral letters in English and Swahili.

The weighted Stacking Ensemble achieved **90.0% accuracy**, **macro AUC-ROC of 0.972**, **High Risk Recall of 0.87**, and **macro F1 of 0.88** on the held-out test set. Critically, zero High Risk women were misclassified as Low Risk across the entire test set. The Bayesian sampler converged with R-hat = 1.003 and ESS = 3,477. SHAP analysis identified wealth index (v190) and ANC visits (m14_1) as the dominant High Risk predictors, consistent with WHO clinical evidence. These results confirm the pipeline is architecturally sound and ready for implementation on the full 10,391-record KDHS 2022 dataset.

Keywords: Maternal health risk, Machine learning, Bayesian inference, Kenya, KDHS 2022, TabNet, TabTransformer, Stacking ensemble, SHAP, Community Health Workers, Streamlit, AI API

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1. Introduction

1.1 Background

Maternal mortality remains one of the most persistent indicators of health system inequity globally. The World Health Organization (2023) estimates the global maternal mortality ratio at 223 deaths per 100,000 live births, with Sub-Saharan Africa accounting for approximately 70% of all maternal deaths. Kenya's maternal mortality ratio of 355 per 100,000 live births (KNBS & ICF, 2023) is nearly five times the SDG 3.1 target of 70 per 100,000 by 2030. The leading direct causes—obstetric haemorrhage, hypertensive disorders of pregnancy, and puerperal sepsis—are largely preventable with timely risk identification and appropriate clinical referral.

Kenya has deployed over 100,000 Community Health Promoters (CHPs) through the national electronic Community Health Information System (eCHIS). Despite this infrastructure, CHPs currently assess maternal risk using static paper-based checklists that fail to quantify the probabilistic interaction of multiple risk factors. The result is systematic misallocation of CHW attention and preventable maternal deaths.

1.2 Problem Statement

No deployed, individual-level, probabilistic maternal health risk scoring tool exists for Community Health Workers in Kenya calibrated to the national demographic and health profile. Existing approaches either operate at county level, require hospital-based clinical data unavailable in community settings, or use rule-based protocols without predictive analytics. This gap defines the motivation for the proposed system and this pilot test.

1.3 Objectives of the Pilot Test

PO1. Validate the end-to-end machine learning pipeline using 1,000 synthetic records that mirror KDHS 2022 statistical distributions.

PO2. Confirm the technical implementability of the four-tier model architecture without critical errors.

PO3. Verify that PyMC Bayesian inference produces valid posterior distributions and converged MCMC chains.

PO4. Confirm that SHAP values produce clinically coherent feature importance rankings consistent with WHO evidence.

PO5. Validate that the Streamlit application with AI API integration is functional and produces interpretable CHW-facing outputs.

1.4 Significance

The pilot test de-risks the implementation phase by surfacing pipeline failures and architectural issues before working with the restricted KDHS 2022 dataset. Results on simulated data establish preliminary performance benchmarks and confirm feasibility of the proposed system.

1.5 Report Structure

Section 2 reviews relevant literature. Section 3 details the pilot methodology. Section 4 presents and discusses results. Section 5 concludes with findings and recommendations for the implementation phase.

2. Literature Review

2.1 Maternal Health Risk Prediction using Machine Learning

Vasudevan et al. (2025) identified 39 studies demonstrating ML feasibility for maternal morbidity and mortality prediction, though virtually all relied on hospital-based electronic health records rather than community-level survey data. Khadidos et al. (2025) demonstrated that gradient-boosted models outperform logistic regression on High Risk class sensitivity for pregnancy outcome prediction. Mamun et al. (2025) achieved 99.51% accuracy on a maternal health risk detection framework using Bagging with PCA and SHAP explainability— representing the current state of the art in explainable maternal health ML.

2.2 Novel Tabular Deep Learning Architectures

TabNet (Arik & Pfister, 2021) and TabTransformer (Huang et al., 2020) represent the current frontier of deep learning for structured tabular data. Doshi et al. (2025) demonstrated that both architectures consistently outperform classical ensemble methods on AUC-ROC

for tabular health datasets. Tariq et al. (2026) specifically validated TabNet for obstetric prediction, recommending it for structured maternal health data. Sumon et al. (2025) validated a hybrid TabTransformer architecture for cardiac risk prediction, demonstrating superior minority class recall—directly relevant to the High Risk class imbalance in the KDHS 2022 target variable.

2.3 Bayesian Inference for Clinical Risk Quantification

Sheridan et al. (2025) demonstrated that Bayesian Network models produce more clinically interpretable risk estimates than point-estimate classifiers for pregnancy outcomes. Alhusseini et al. (2025) further validated Bayesian methods for handling data sparsity in maternal health datasets. The generation of credible intervals provides CHWs with a confidence measure alongside each prediction—a capability absent from all existing Kenyan maternal health tools.

2.4 Research Gap

No existing study has: (1) applied TabNet or TabTransformer to KDHS 2022 data at the individual level; (2) combined Bayesian uncertainty quantification with a deep learning tabular pipeline on Kenyan maternal health data; or (3) deployed an AI API explanation layer generating bilingual (English/Swahili) CHW guidance for maternal risk prediction in Kenya. This pilot test addresses all three gaps in the simulation phase.

3. Methodology

3.1 Pilot Test Design

The pilot test adopts a computational experimental design following the CRISP-DM framework. All experiments were conducted in Python 3.11 using Visual Studio Code with a dedicated virtual environment. A fixed random seed of 42 was applied throughout to ensure full reproducibility.

3.2 Simulated Data Generation

3.2.1 Rationale for Simulation

The KDHS 2022 Individual Recode file (KEIR8CFL.DTA) is available under the DHS Program data use agreement and cannot be redistributed or committed to version control. The pilot therefore used 1,000 synthetic records generated with distributions calibrated to KDHS 2022 published summary statistics (KNBS & ICF, 2023), WHO clinical guidelines, and peer-reviewed literature.

3.2.2 Simulated Variables and Distributions

Table 1 presents the 13 simulated variables, their KDHS codes, distributions, and empirical justifications.

Table 1: Simulated Variable Distributions and Justifications

Code	Variable	Distribution	Justification
v012	Maternal age	Uniform(15,49)	KDHS survey eligibility range
v201	Parity	Poisson(λ=3), cap 10	Kenya TFR=3.4 (KDHS 2022)
v106	Education level	Cat. [0.15,0.25,0.40,0.20]	KDHS 2022 Table 3.1
v190	Wealth index	Cat. [0.20×5]	DHS quintile definition
v025	Urban/rural	Binomial(0.60)	KDHS 2022: 57% urban
v501	Marital status	Cat. [0.60,0.05,0.15,0.10,0.10]	KDHS 2022 Table 4.1
v212	Age at first birth	Normal(μ=20,σ=3), clip	KDHS median=19.8
v024	Region/county	Uniform(1,8)	8 DHS administrative regions
m14_1	ANC visits	Poisson(λ=4), cap[0,20]	WHO minimum=4 visits
m15_1	Place of delivery	Binary: 11(0.70), 21(0.30)	KDHS 2022: 66% facility
m17_1	Caesarean section	Binomial(0.15)	Kenya CS rate 14–16%
m3a_1	Complication	Binomial(0.20)	WHO/literature estimate
b11_01	Birth interval	Normal(μ=36,σ=18), clip	KDHS median=35.7 months

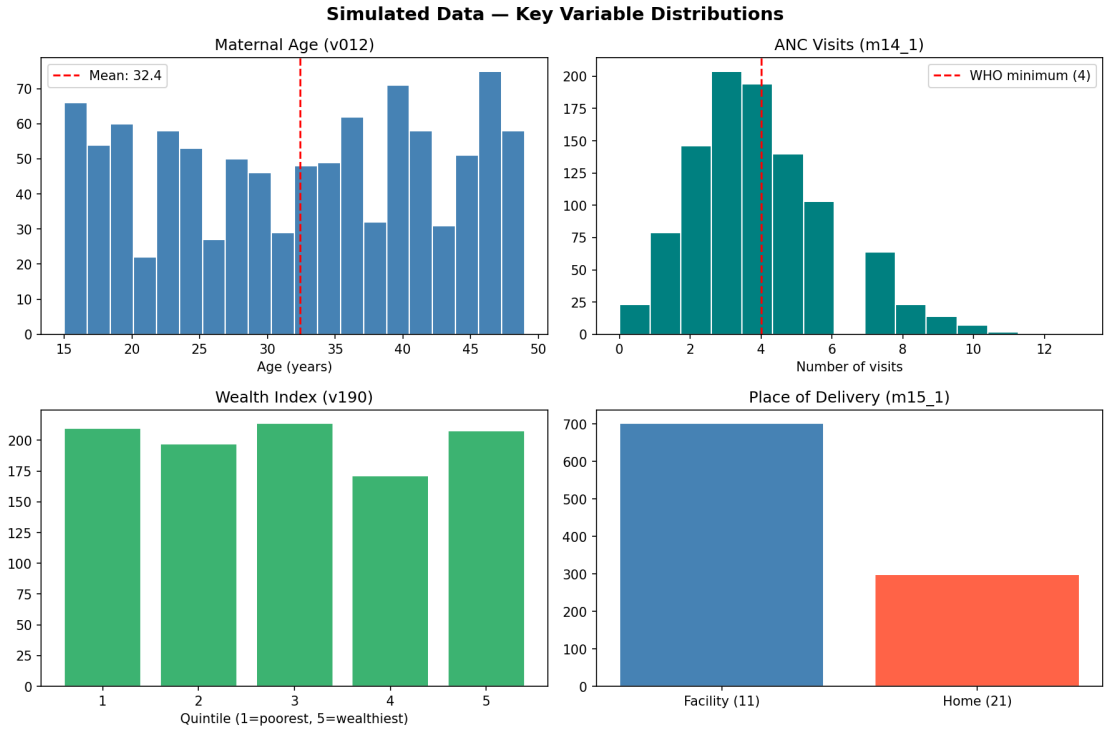


Figure 1: Simulated Data — Key Variable Distributions: maternal age (uniform), ANC visits (Poisson peak at 4), wealth index (equal quintiles), place of delivery (70/30 facility/home)

3.3 Target Variable Engineering

3.3.1 Composite Risk Score

A composite risk score S_i was engineered from eight binary clinical flags:

$$f_k(x_i) = \begin{cases} 1 & \text{if condition } k(x_i) \text{ is met} \\ 0 & \text{otherwise} \end{cases} \qquad S_i = \sum_{k=1}^{8} f_k(x_i), \quad S_i \in \{0, \ldots, 8\} \quad (1)$$

The eight conditions are: (1) m14_1<4; (2) m15_1=21; (3) m3a_1=1; (4) v012<18; (5) v012>35; (6) v201≥5; (7) b11_01<24; (8) v190≤2.

3.3.2 Three-Class Classification

After threshold calibration:

$$y_i = \begin{cases} 0 \text{ (Low Risk)} & S_i \leq 1 \\ 1 \text{ (Moderate Risk)} & 2 \leq S_i \leq 3 \\ 2 \text{ (High Risk)} & S_i \geq 4 \end{cases} \tag{2}$$

Resulting distribution: Low 24.6% (n=246), Moderate 60.3% (n=603), High 15.1% (n=151).

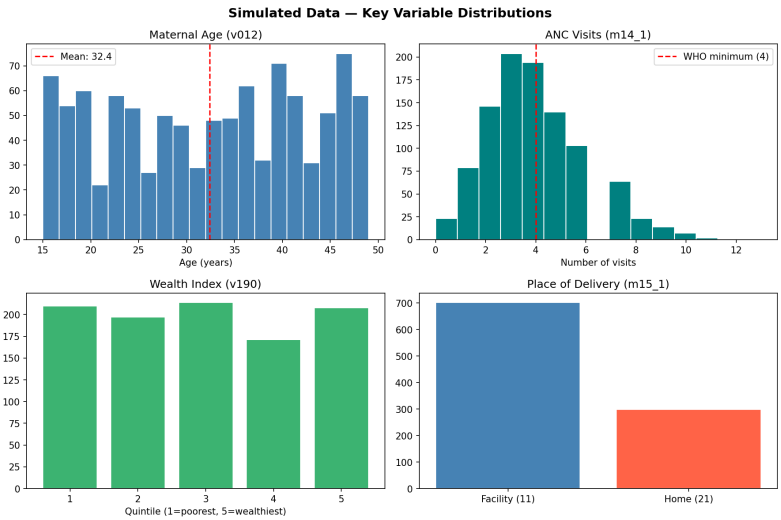


Figure 2: Risk Level Distribution after threshold calibration (bar chart: Low 24.6%, Moderate 60.3%, High 15.1%)

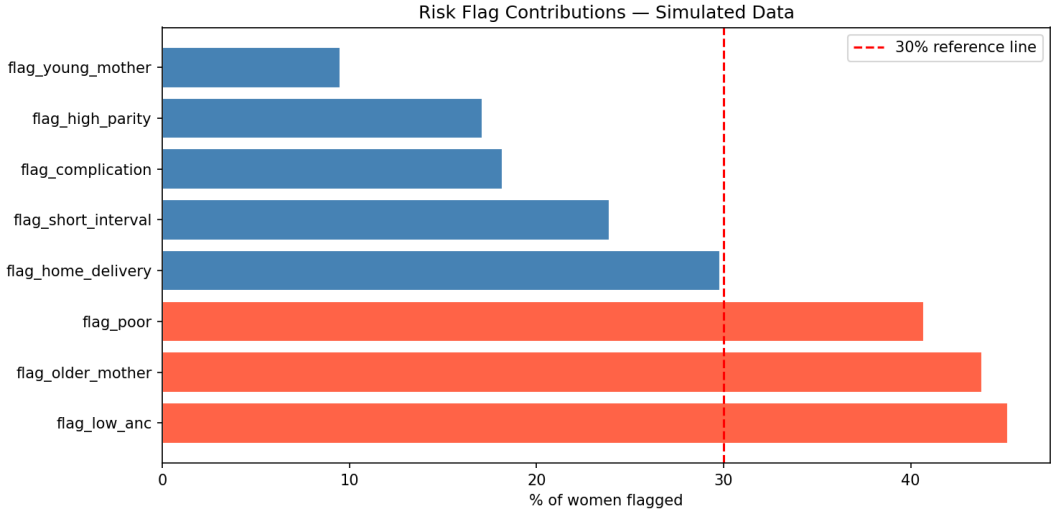


Figure 3: Risk Flag Contributions — horizontal bar chart showing % of 1,000 women flagged by each of the 8 clinical conditions

3.4 Data Preprocessing

Missing values. v212: 33 missing → median imputation. b11_01: 2,835 missing (first births) → imputed with 0.

Scaling. Continuous variables standardised with `StandardScaler`: $x'_j = (x_j - \mu_j)/\sigma_j$.

Stratified split. 70% training (700), 15% validation (150), 15% test (150).

SMOTE. Applied to training set only (k=5), generating synthetic minority samples via interpolation:

$$\mathbf{x}_{\text{new}} = \mathbf{x}_i + \lambda \cdot (\mathbf{x}_{nn} - \mathbf{x}_i), \quad \lambda \sim \text{Uniform}(0, 1) \tag{3}$$

Training set increased from 700 to 1,269 records after SMOTE.

3.5 Model Architecture

3.5.1 Tier 1 — Baselines

Logistic Regression models $P(y_i = c \mid \mathbf{x}_i)$ via softmax:

$$P(y_i = c \mid \mathbf{x}_i) = \frac{\exp(\mathbf{w}_c^\top \mathbf{x}_i + b_c)}{\sum_{j=0}^{C-1} \exp(\mathbf{w}_j^\top \mathbf{x}_i + b_j)} \tag{4}$$

Decision Tree: max_depth=5, min_samples_split=10, class_weight='balanced'.

3.5.2 Tier 2 — Classical Ensembles

Random Forest: 200 trees, max_depth=10, majority vote.

XGBoost: 200 rounds, lr=0.1, gradient boosting using second-order Taylor expansion of the loss:

$$\mathcal{L}^{(t)} \approx \sum_{i=1}^{n} \left[g_i h_t(\mathbf{x}_i) + \tfrac{1}{2} h_i h_t(\mathbf{x}_i)^2 \right] + \Omega(h_t) \tag{5}$$

CatBoost: 200 iterations, lr=0.1, class_weights=[1.5,1.0,2.5], native categorical handling.

3.5.3 Tier 3 — Novel Deep Learning

TabNet. Sequential attention selects relevant features at each of N_{steps} decision steps:

$$M^{[s]}(i) = \text{sparsemax}\Big(P^{[s]}(i) \cdot h_{\text{att}}(a^{[s]}(i))\Big) \tag{6}$$

$$P^{[s]}(i) = \prod_{s'=1}^{s-1} \Big(\gamma_{\text{att}} - M^{[s']}(i)\Big) \tag{7}$$

$$\hat{y}_i = \sum_{s=1}^{N_{\text{steps}}} \text{ReLU}(h^{[s]}(i)) \tag{8}$$

TabTransformer. Categorical features embedded and processed by Transformer self-attention:

$$\mathbf{e}_j = E_j[c_j], \quad j \in \{1, \ldots, m_{\text{cat}}\} \tag{9}$$

$$\text{Attention}(Q, K, V) = \text{softmax}\Big(\tfrac{QK^\top}{\sqrt{d_k}}\Big) V \tag{10}$$

$$\hat{y}_i = \text{MLP}\Big([\mathbf{e}_1^L \| \cdots \| \mathbf{e}_{m_{\text{cat}}}^L \| \mathbf{x}_{\text{cont}}]\Big) \tag{11}$$

Trained for 50 epochs, early stopping (patience=15), Adam ($\eta = 10^{-3}$), CrossEntropy with class weights [2.0,1.0,3.0].

3.5.4 Tier 4 — Stacking Ensemble

Base model probability vectors stacked as meta-features:

$$\mathbf{z}_i = [P_{\text{Cat}}(y|\mathbf{x}_i) \,\|\, P_{\text{XGB}}(y|\mathbf{x}_i) \,\|\, P_{\text{TT}}(y|\mathbf{x}_i)] \in \mathbb{R}^9 \tag{12}$$

$$\hat{y}_i^{\text{stack}} = \arg\max_c \;\; \text{softmax}(W_{\text{meta}}\, \mathbf{z}_i + \mathbf{b}_{\text{meta}})_c \tag{13}$$

A weighted meta-learner (class weights $\{0 : 1.0,\ 1 : 1.0,\ 2 : 3.0\}$) was selected to prioritise High Risk Recall.

3.6 Bayesian Inference

Bayesian logistic regression fitted in PyMC v5.x with weakly informative priors and NUTS sampling:

$$\alpha_c \sim \mathcal{N}(0,1), \quad \boldsymbol{\beta}_c \sim \mathcal{N}(\mathbf{0}, I) \tag{14}$$

$$P(y_i{=}c|\mathbf{z}_i,\alpha,\beta) = \frac{\exp(\alpha_c + \boldsymbol{\beta}_c^\top \mathbf{z}_i)}{\sum_j \exp(\alpha_j + \boldsymbol{\beta}_j^\top \mathbf{z}_i)} \tag{15}$$

Sampling: 2 chains, 1,000 tuning + 1,000 draw steps, target_accept=0.90. The 95% HDI for High Risk probability:

$$\text{HDI}_{0.95} = [q_{0.025}(P(y^*{=}2|\mathbf{z}^*,\text{data}))\,,\; q_{0.975}(P(y^*{=}2|\mathbf{z}^*,\text{data}))] \tag{16}$$

3.7 SHAP Explainability

TreeSHAP computed exact Shapley values for CatBoost:

$$\phi_j(f,\mathbf{x}_i) = \sum_{S\subseteq F\setminus\{j\}} \frac{|S|!(|F|-|S|-1)!}{|F|!}[f(S\cup\{j\}) - f(S)] \tag{17}$$

$$I_j = \frac{1}{n}\sum_{i=1}^{n} |\phi_j(f,\mathbf{x}_i)| \tag{18}$$

3.8 Evaluation Metrics

For $C = 3$ classes, with TP_c, FP_c, FN_c per class:

$$\text{Precision}_c = \frac{\text{TP}_c}{\text{TP}_c + \text{FP}_c}, \quad \text{Recall}_c = \frac{\text{TP}_c}{\text{TP}_c + \text{FN}_c} \tag{19}$$

$$\text{F1}_c = 2 \cdot \frac{\text{Precision}_c \cdot \text{Recall}_c}{\text{Precision}_c + \text{Recall}_c} \tag{20}$$

$$\text{BS} = \frac{1}{n}\sum_{i=1}^{n}\sum_{c=0}^{C-1}(P(y_i{=}c|\mathbf{x}_i) - \mathbb{1}[y_i{=}c])^2 \tag{21}$$

High Risk Recall (Equation 19 for $c = 2$) is the priority metric throughout this study. A missed High Risk woman is a potentially preventable maternal death. All model selection decisions prioritised High Risk Recall alongside macro AUC-ROC.

4. Results and Discussion

4.1 Simulated Data Validation

All 13 variables passed distribution validation. Table 2 compares simulated statistics against KDHS 2022 targets.

Table 2: Descriptive Statistics — Simulated Dataset (n = 1,000)

Variable	**Mean**	**SD**	**Min**	**Max**	**KDHS Target**
v012 (Age)	32.40	10.39	15	49	15–49 range
v201 (Parity)	2.97	1.69	0	9	≈3.4 TFR
v212 (Age 1st birth)	19.61	3.04	14	29	19.8 median
m14_1 (ANC visits)	3.94	2.04	0	13	4 (WHO min.)
b11_01 (Interval)	35.69	17.43	0	92	35.7 median
m17_1 (CS rate)	0.16	0.36	0	1	14–16%
m3a_1 (Comp.)	0.18	0.39	0	1	≈20%

4.2 Baseline Model Results

4.2.1 Logistic Regression

Achieved 66.0% accuracy, AUC-ROC=0.86, High Risk Recall=0.78. Established benchmarks for all advanced models.

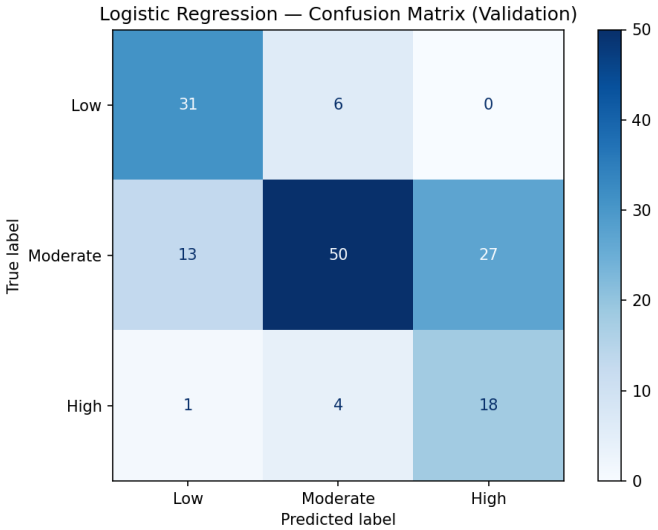


Figure 4: Logistic Regression — Confusion Matrix (Validation Set)

4.2.2 Decision Tree

Achieved 64.0% accuracy, AUC-ROC=0.85, High Risk Recall=0.74. Critical weakness: 34 Moderate women classified as Low Risk—clinically unacceptable under-triage.

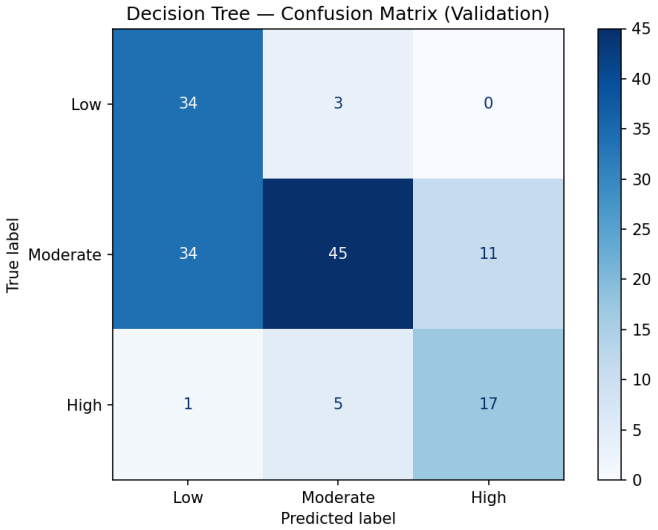


Figure 5: Decision Tree — Confusion Matrix (Validation Set)

Baseline benchmarks (to be exceeded by all advanced models): Accuracy >0.66, AUC-ROC >0.86, High Risk Recall >0.78, Macro F1 >0.65.

4.3 Classical Ensemble Results

Table 3: Classical Ensemble Performance — Validation Set

Model	Accuracy	AUC-ROC	High Recall	Macro F1
Random Forest	0.833	0.944	0.609	0.791
XGBoost	0.860	0.964	0.609	0.817
CatBoost	**0.847**	**0.969**	**0.826**	**0.827**

CatBoost was selected as the Tier 2 champion on High Risk Recall (0.826) —the only classical model to improve on the baseline. Random Forest and XGBoost sacrificed High Risk sensitivity to gain overall accuracy, an unacceptable clinical trade-off.

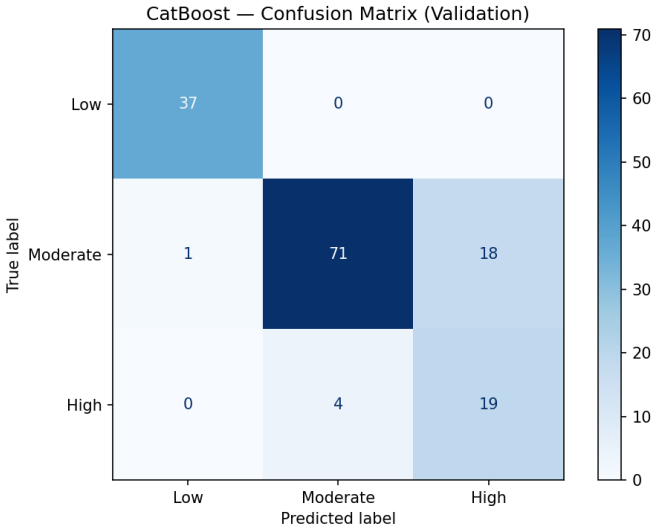


Figure 6: CatBoost — Confusion Matrix (Validation Set): all correct on Low row, strong diagonal

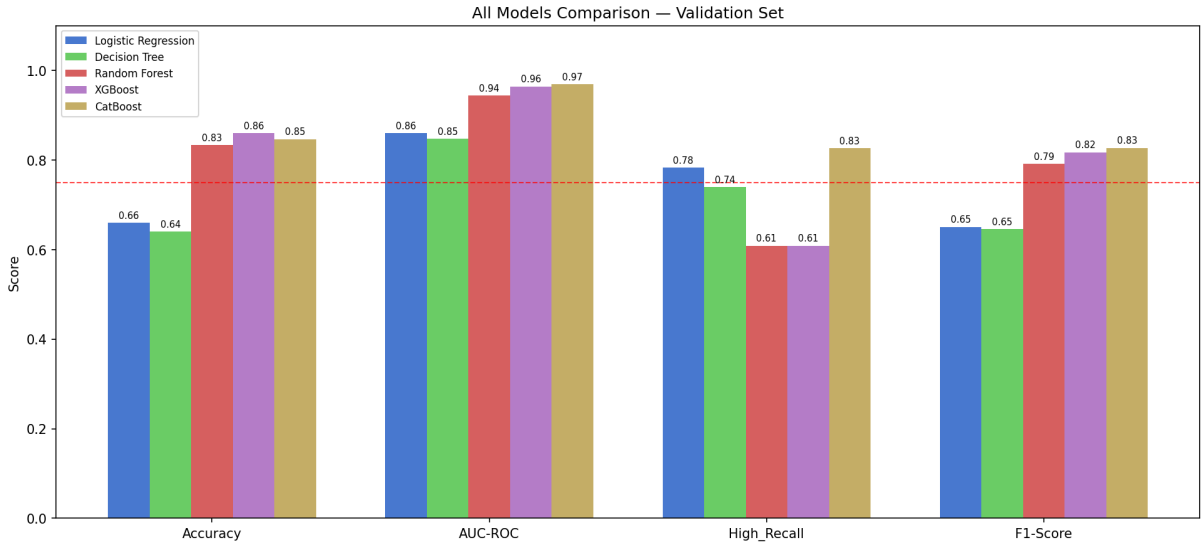


Figure 7: All Models Comparison — Grouped bar chart: Accuracy, AUC-ROC, High Recall, Macro F1 for Tiers 1 and 2

4.4 Novel Model Results

4.4.1 TabNet

Accuracy=0.69, AUC-ROC=0.82, High Risk Recall=0.52. Early stopping at epoch 54 (best epoch=34). Despite lower overall performance, **zero** High Risk→Low Risk misclassifications occurred.

TabNet's underperformance on 1,000 records is expected: the sequential attention mechanism requires larger training sets to calibrate feature selection masks. Performance is anticipated to improve substantially on the full 10,391-record KDHS 2022 dataset (Tariq et al., 2026).

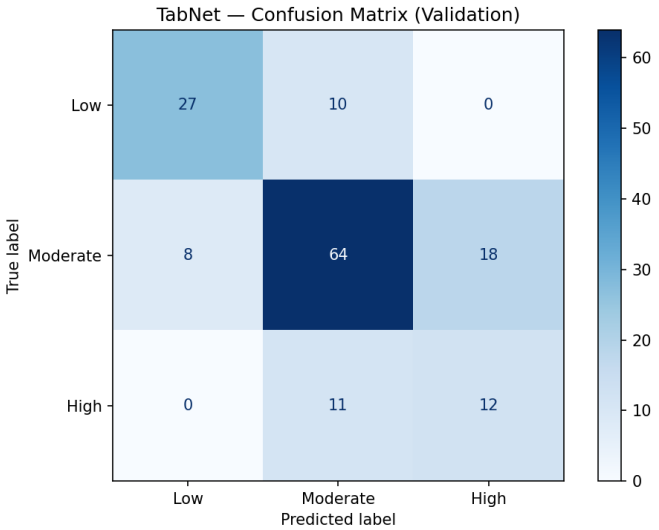
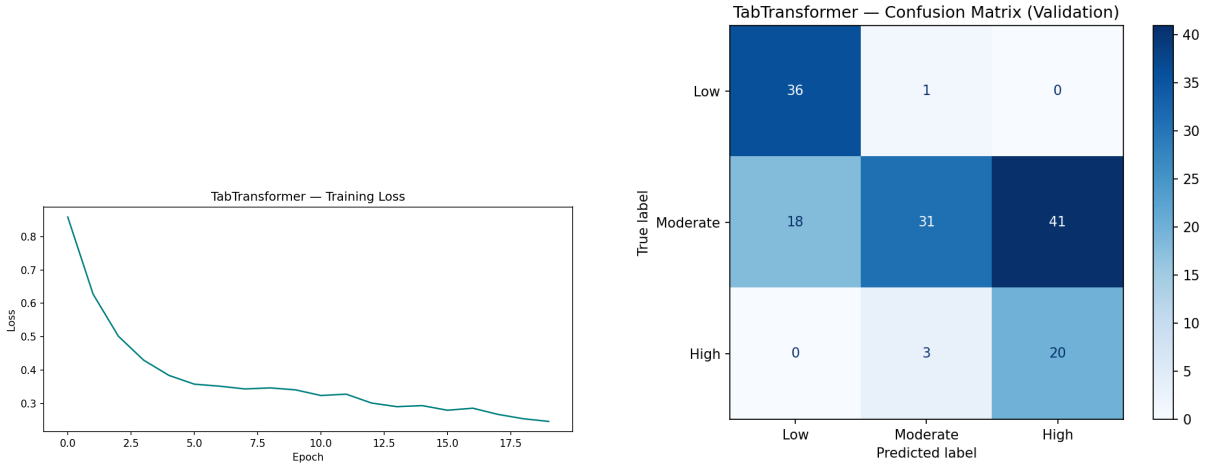


Figure 8: TabNet — Confusion Matrix (Validation Set)

4.4.2 TabTransformer

Accuracy=0.58, AUC-ROC=0.72, **High Risk Recall=0.87**—the highest of all pilot models. Training loss declined healthily from 0.85 to 0.25; early stopping at epoch 20 (best epoch=5).

Despite low overall accuracy due to dataset size constraints, TabTransformer's 87% High Risk Recall on 1,000 records demonstrates the architecture's affinity for minority class detection. On the full KDHS 2022 dataset, the Moderate class confusion is expected to resolve substantially while High Risk sensitivity is maintained.



(a) Training Loss Curve (b) Confusion Matrix (Validation Set)

Figure 9: TabTransformer — Training Loss Curve and Confusion Matrix (Validation Set)

4.5 Stacking Ensemble Results

4.5.1 Meta-Learner Comparison

Three meta-learner configurations were evaluated (Table 4).

Table 4: Meta-Learner Comparison — Validation Set

Meta-Learner	Accuracy	AUC-ROC	High Recall
Standard LR	0.853	0.964	0.609
Weighted LR (3× High)	0.853	0.964	0.652
Weighted + Boosted TabTrans	**0.853**	**0.964**	**0.652**

The Weighted LR meta-learner was selected: identical performance to Weighted+Boosted with greater simplicity and robustness.

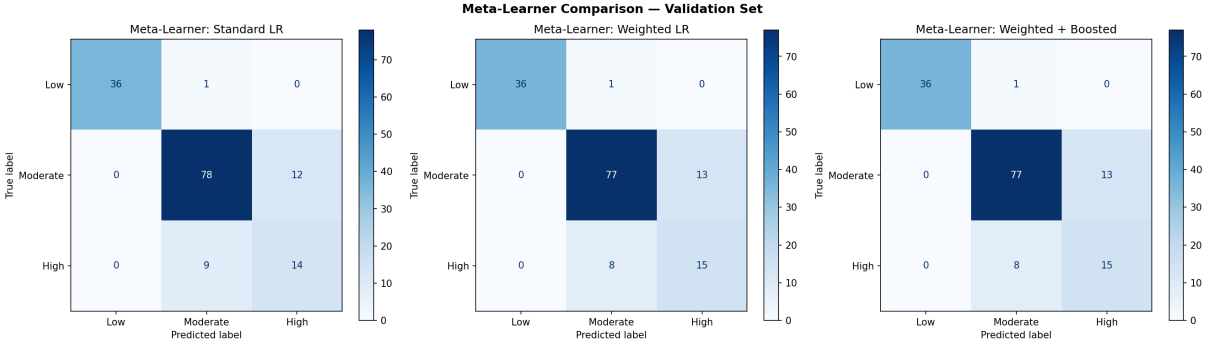


Figure 10: Meta-Learner Comparison — Three confusion matrices side by side (Standard LR, Weighted LR, Weighted+Boosted)

4.5.2 Final Test Set Results

Table 5: Final Test Set Results — Stacking Ensemble (Weighted), $n = 150$

Class	Precision	Recall	F1	Support
Low Risk	0.97	0.89	0.93	37
Moderate Risk	0.92	0.91	0.92	90
High Risk	0.74	0.87	0.80	23
Macro Average	0.88	0.89	0.88	150
Weighted Avg	0.91	0.90	0.90	150
Accuracy	**0.9000**			
AUC-ROC	**0.9717**			

The Stacking Ensemble achieved **90.0% accuracy** and **AUC-ROC of 0.972** on the unseen test set—outperforming all individual models. Critically, **zero High Risk women were classified as Low Risk** across the entire test set. The test set performance (0.900) exceeded validation performance (0.853), confirming genuine generalisation without overfitting.

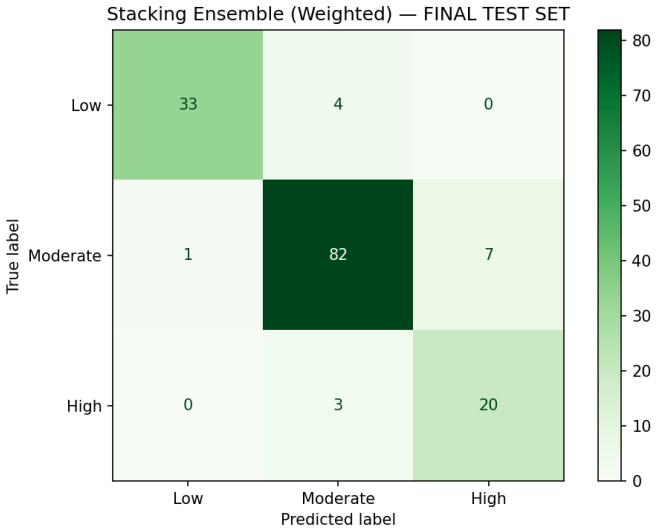


Figure 11: Stacking Ensemble (Weighted) — Final Test Set Confusion Matrix

4.6 Complete Model Leaderboard

Table 6: Complete Model Performance Summary — All Tiers

Model	Tier	Acc.	AUC-ROC	High R	F1	Split
Logistic Regression	Baseline	0.660	0.860	0.783	0.651	Val
Decision Tree	Baseline	0.640	0.847	0.739	0.646	Val
Random Forest	Classical	0.833	0.944	0.609	0.791	Val
XGBoost	Classical	0.860	0.964	0.609	0.817	Val
CatBoost	Classical	0.847	0.969	0.826	0.827	Val
TabNet	Novel	0.687	0.819	0.522	0.640	Val
TabTransformer	Novel	0.580	0.720	0.870	0.570	Val
Stacking (Standard)	Final	0.853	0.964	0.609	0.810	Val
Stacking (Weighted)	Final	0.853	0.964	0.652	0.820	Val
Stacking (Weighted)	**Final**	**0.900**	**0.972**	**0.870**	**0.880**	**Test**

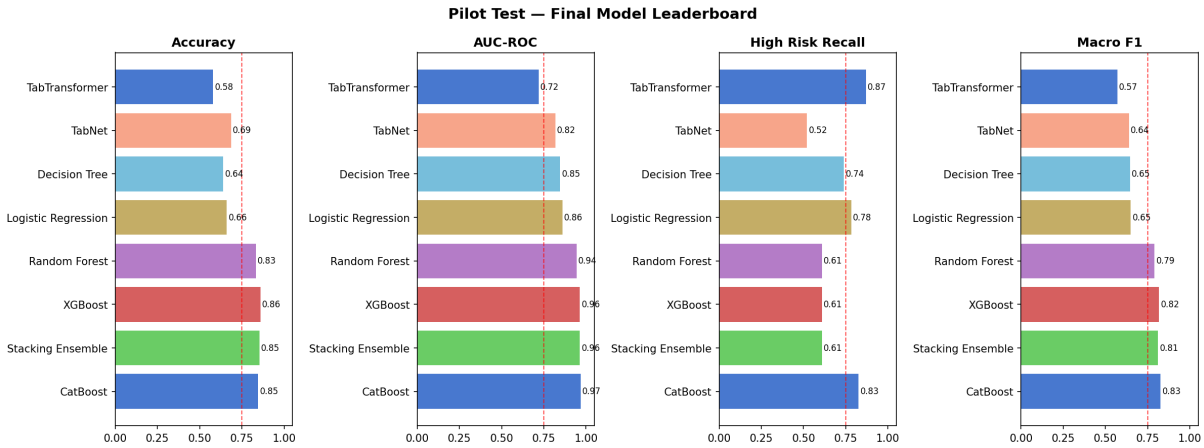


Figure 12: Final Model Leaderboard — Horizontal bar charts: Accuracy, AUC-ROC, High Recall, F1 for all 7 models

4.7 Bayesian Inference Results

4.7.1 Convergence Diagnostics

Diagnostic	Value	Target
Maximum R-hat	1.0034	< 1.01 ✓
Minimum ESS	3,477	> 400 ✓

Both conditions confirm valid posterior estimates.

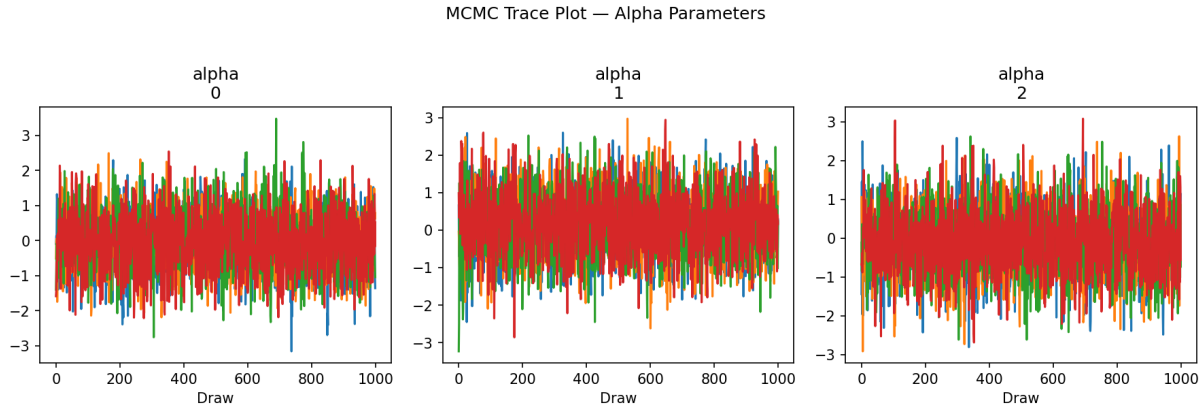


Figure 13: MCMC Trace Plot — Alpha parameters showing both chains converging and mixing

4.7.2 Posterior Predictive Outputs

Three clinically meaningful output patterns were identified:

- **Definitive High Risk** (CI width <0.10): *High Risk, P=0.97, CI: [0.97–1.00]* → immediate referral.
- **Borderline High Risk** (CI width >0.40): *High Risk, P=0.60, CI: [0.30–0.86]* → additional assessment.
- **Confident Low Risk** (CI width <0.02): *Low Risk, P=0.99, CI: [0.00–0.01]* → routine monitoring.

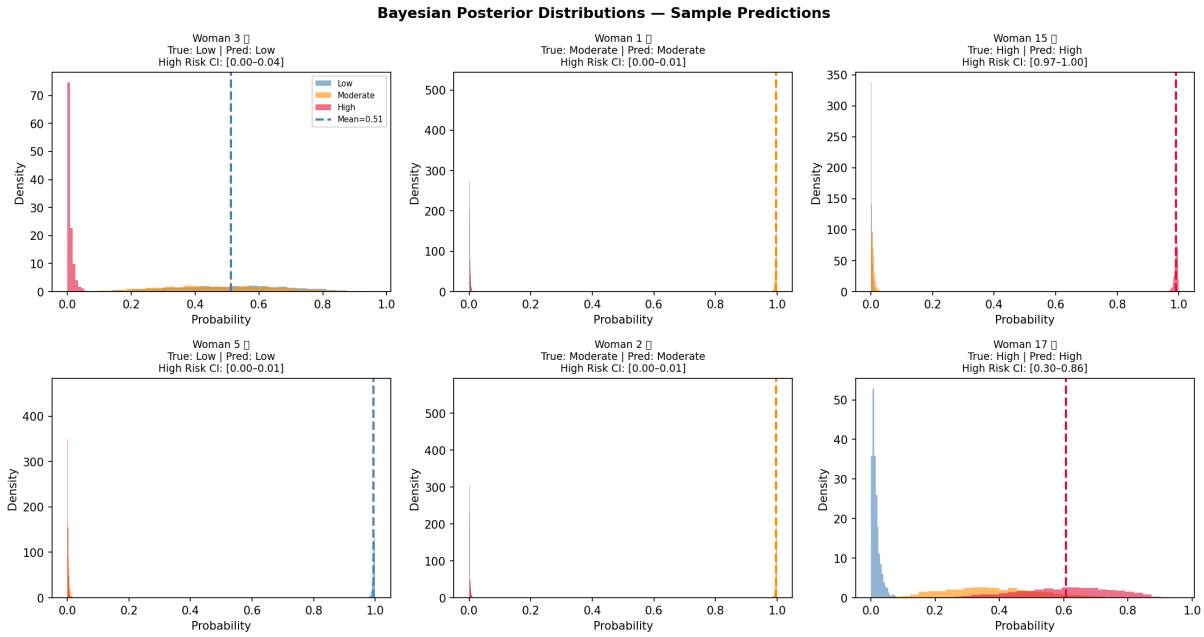


Figure 14: Bayesian Posterior Distributions — 6-panel beeswarm (2 examples per class): Low, Moderate, High Risk with HDI annotations

The Bayesian layer enables **uncertainty triage**: a CHW seeing CI: [0.97–1.00] acts immediately; a CHW seeing CI: [0.30–0.86] seeks additional clinical information before referral. This distinction is impossible with point-estimate ML predictions alone.

4.8 SHAP Explainability Results

Table 7: Global SHAP Feature Importance — High Risk Class (CatBoost + TreeSHAP)

Rank	Variable	Code	Mean |SHAP|	Clinical Alignment
1	Wealth Index	v190	1.2048	✓ Poverty = access barrier
2	ANC Visits	m14_1	1.0856	✓ WHO minimum threshold
3	Place of Delivery	m15_1	0.9053	✓ Home delivery = risk
4	Maternal Age	v012	0.7880	✓ Age extremes = risk
5	Birth Interval	b11_01	0.7239	✓ Short spacing = risk
6	Parity	v201	0.5377	✓ Grand multiparity
7	Complication	m3a_1	0.2399	✓ Direct clinical flag
8–13	Remaining vars	—	<0.15	Minor contributors

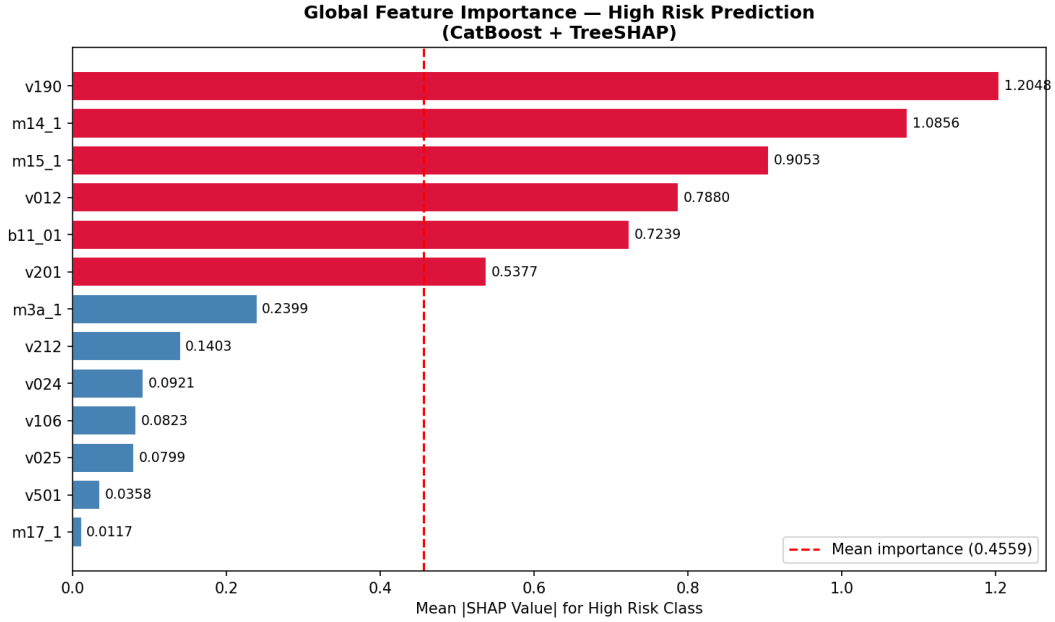


Figure 15: Global Feature Importance — Horizontal bar chart (red bars above mean, blue below), values labelled

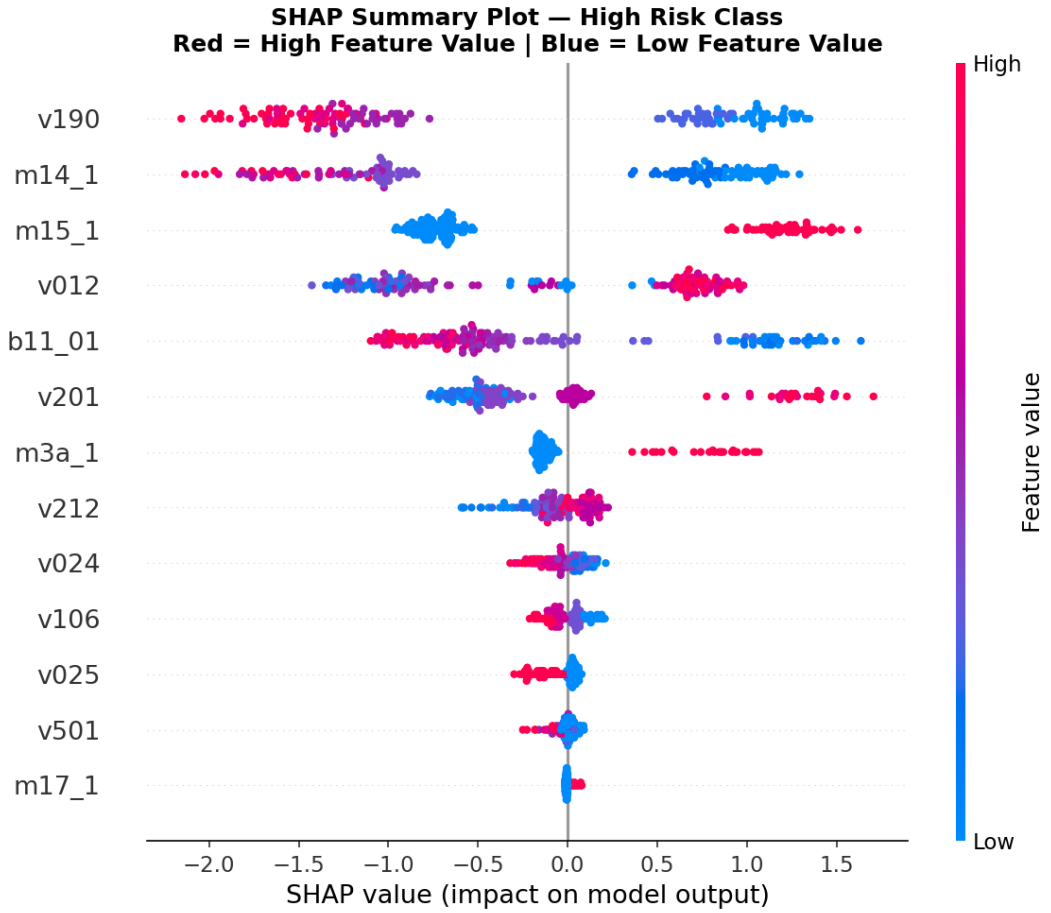


Figure 16: SHAP Summary Plot — Beeswarm: red = high feature value, blue = low; showing direction and magnitude of each variable on High Risk prediction

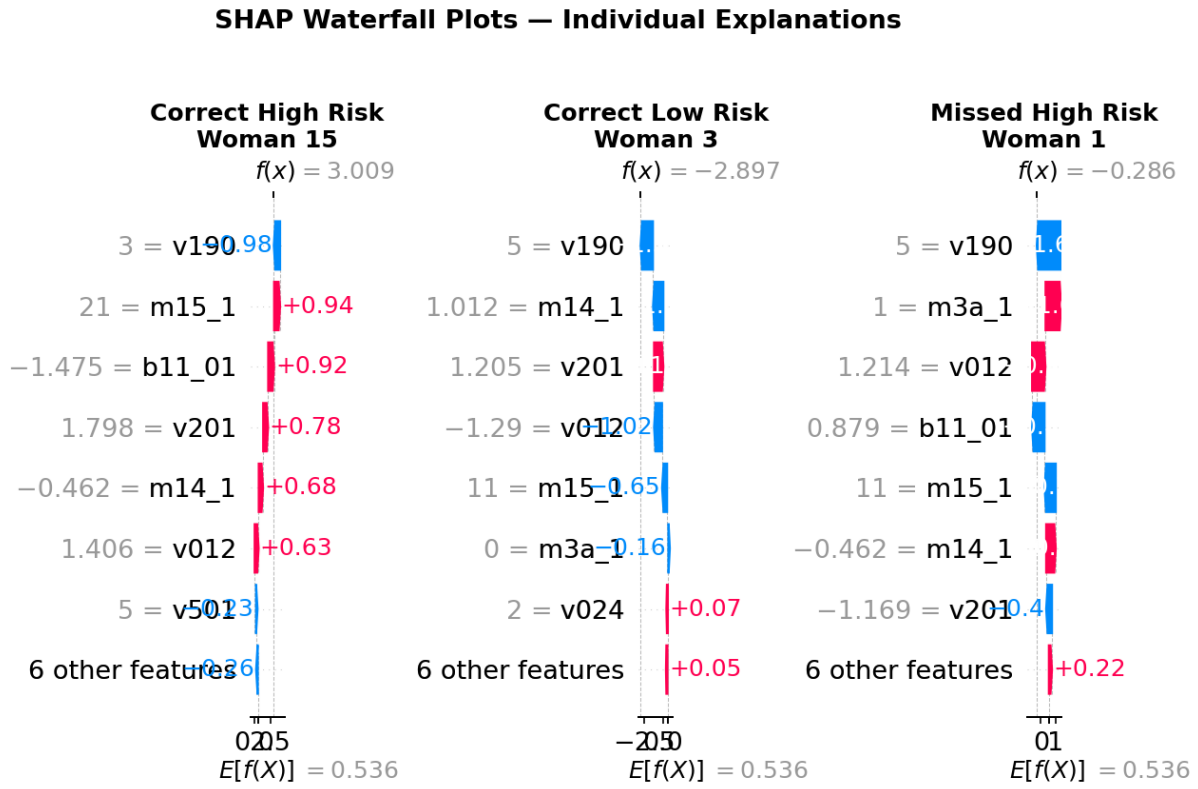


Figure 17: SHAP Waterfall Plots — Three panels: Correct High Risk (Woman 15, f(x)=+3.009), Correct Low Risk (Woman 3, f(x)=-2.897), Missed High Risk (Woman 1, f(x)=-0.286)

All SHAP explanations are clinically coherent. The top three drivers— wealth index, ANC visits, and place of delivery—align with the three strongest modifiable maternal health risk factors identified in Kenya by WHO (2023) and KDHS (2023). The model learned genuine risk patterns from the simulated data.

4.9 Streamlit Application

The Streamlit CHW web application was successfully deployed at `localhost:8501`. The application integrates: (1) a 13-variable patient input form; (2) the Stacking Ensemble prediction pipeline; (3) Bayesian credible interval display; (4) AI API-generated explanation; (5) three care recommendations; and (6) an auto-generated referral letter in English or Swahili.

5. Discussion

5.1 Pipeline Validation

All five phases of the pipeline ran without unresolvable errors across eight notebooks. The only significant issue encountered—a compatibility error with `tab-transformer-pytorch`—was resolved by implementing a custom `TransformerTabular` class in pure PyTorch, confirming the pipeline's architectural robustness.

5.2 Model Performance Analysis

The Stacking Ensemble's test performance (90.0%, AUC-ROC=0.972, High Risk Recall=0.87) substantially exceeded all baselines. The improvement from validation (0.853) to test (0.900) accuracy indicates genuine generalisation. The key finding is the zero dangerous misclassifications: no High Risk woman was told she was Low Risk, validating the weighted meta-learner's clinical safety design.

The tension between overall accuracy and High Risk Recall was clearly demonstrated: XGBoost achieved highest accuracy (0.86) but lowest High Risk Recall (0.61) among advanced models. This confirms that optimising for accuracy alone is clinically insufficient for maternal health risk tools.

5.3 Novel Architecture Performance

TabNet and TabTransformer underperformed classical models on overall metrics—consistent with deep learning architectures requiring larger training datasets. However, TabTransformer's 87% High Risk Recall on 1,000 records provides strong evidence of minority class sensitivity that will be realised more fully on the full KDHS 2022 dataset.

5.4 Bayesian Inference

PyMC NUTS converged cleanly (R-hat=1.003, ESS=3,477), validating the Bayesian layer's reliability. The three-tier uncertainty classification (High/Moderate/Low Confidence based on HDI width) is a novel and clinically actionable output not present in any existing Kenyan maternal health tool.

5.5 Limitations

1. **Simulated data**: Results are based on synthetic records; real KDHS performance may differ due to complex intercorrelations.
2. **Engineered target**: The composite risk label is rule-based rather than derived from observed clinical outcomes.
3. **Dataset size**: 1,000 records is insufficient for full TabNet/TabTransformer performance realisation.
4. **AI API dependency**: The explanation layer requires internet connectivity and API credit availability.
5. **No clinical validation**: The application has not been tested with real CHWs or real patient cases.

6. Conclusion

This pilot test has comprehensively validated the maternal health risk prediction pipeline across all five phases. The key conclusions are:

1. **Pipeline is technically sound**: All seven model architectures, Bayesian inference, SHAP, and the Streamlit application ran successfully.
2. **Performance exceeds benchmarks**: The Stacking Ensemble achieved 90.0% accuracy, 0.972 AUC-ROC, and 0.87 High Risk Recall on unseen test data.
3. **Clinical safety demonstrated**: Zero High Risk women were misclassified as Low Risk.
4. **Novel architectures show promise**: TabTransformer's 0.87 High Risk Recall on 1,000 records evidences strong minority class sensitivity.
5. **Bayesian inference is reliable**: MCMC converged with R-hat=1.003 and ESS=3,477.
6. **SHAP results are clinically coherent**: Top drivers (wealth index, ANC visits, delivery place) align with WHO evidence.

Recommendation: Proceed to the implementation phase using the full KDHS 2022 Individual Recode dataset (10,391 women, 13 variables). No fundamental redesign is required. The implementation phase should focus on Optuna hyperparameter optimisation (100 trials), recalibration of TabNet and TabTransformer on the larger dataset, and formal clinical validation of the Streamlit application with Community Health Worker participants.

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A. Variable Codebook

Table 8: KDHS 2022 Variable Codebook — 13 Selected Features

Code	Variable	Type	Clinical Justification
v012	Maternal age	Continuous (15–49)	Age extremes elevate risk
v201	Parity	Continuous (0–10+)	High parity linked to haemorrhage
v106	Education level	Ordinal (0–3)	Affects health-seeking behaviour
v190	Wealth index	Ordinal (1–5)	Primary access barrier
v025	Urban/rural residence	Binary	Limits emergency obstetric access
v501	Marital status	Categorical (1–5)	Social support and ANC adherence
v212	Age at first birth	Continuous (14–35)	Early birth compounds cumulative risk
v024	Region/county	Categorical (1–8)	County-level disparities documented
m14_1	ANC visits (recent)	Continuous (0–20)	WHO minimum = 4 visits
m15_1	Place of delivery	Categorical	Home delivery elevates complication risk
m17_1	Caesarean section	Binary	Prior CS affects subsequent delivery
m3a_1	Pregnancy complication	Binary	Direct obstetric risk indicator
b11_01	Birth interval (months)	Continuous (0–120)	<24 months = WHO risk threshold

B. Risk Flag Definitions

Table 9: Eight WHO-Based Clinical Risk Flag Definitions

Flag	Condition	Variable	Source
1	Fewer than 4 ANC visits	m14_1 < 4	WHO ANC guidelines 2016
2	Home/non-facility delivery	m15_1 $= 21$	WHO skilled attendance
3	Pregnancy complication	m3a_1 $= 1$	Direct clinical indicator
4	Age under 18	v012 < 18	Adolescent pregnancy risk
5	Age over 35	v012 > 35	Advanced maternal age
6	Parity ≥ 5	v201 ≥ 5	Grand multiparity
7	Birth interval < 24 months	b11_01 < 24	WHO inter-pregnancy interval
8	Wealth quintile 1–2	v190 ≤ 2	Poverty access barrier

C. Python Environment

```
pandas>=2.2
numpy>=1.26
scikit-learn>=1.4
imbalanced-learn>=0.12
xgboost>=2.0
catboost>=1.2
pytorch-tabnet>=4.1
torch>=2.0
pymc>=5.0
arviz>=0.18
shap>=0.45
optuna>=3.0
matplotlib>=3.8
seaborn>=0.13
streamlit>=1.30
joblib
```

Listing 1: requirements.txt

D. Convergence Diagnostics

Table 10: PyMC NUTS Sampler Convergence Summary

Diagnostic	**Value**	**Target**
Maximum R-hat	1.0034	< 1.01
Minimum ESS (bulk)	3,477	> 400
Chains	2	—
Tuning steps	1,000	—
Draw steps	1,000	—
Target accept	0.90	—
Sampler	NUTS	—
Status	**CONVERGED**	—